

Forecasting without quality data? That's like driving blindfolded!

Demand Planning app – Data Import



In this second article in my series on Demand Planning, we take a close look at what it takes to import high-quality data into the Demand Planning app in Dynamics 365 Supply Chain Management.

From system setup to import profiles — and from data quality pitfalls to practical setup — this guide will help you lay a strong foundation for accurate, AI-powered forecasts.

Great forecasting doesn't start with models. It starts with data!

Imagine you're standing at the beginning of an important mission. Your goal is to optimize your company's forecasting and overall supply chain planning.

Microsoft Dynamics 365 Demand Planning is designed to help you achieve that. Using AI, machine learning, and a no-code interface, it puts the power of advanced forecasting in your hands. But even the smartest forecast engine can't work its magic without one main ingredient: **high-quality data**.

In fact, studies revealed by Gardner in 2021 showed that 88% of companies struggle with poor data quality, leading to an average 12% loss in revenue.

So, when you're getting ready to use Demand Planning, remember your journey starts not with the forecasts, but with the **data import**.

Before you import your data, take a moment and consider - is your data telling a coherent story?

Demand planning relies on raw data, which is converted into time series to predict future demand. Inputs such as unit sales by product or location over a specific period are used in the forecasting calculations. Product-related information adds further context to the time series, helping users better understand and interpret the results.

The starting point for data import is understanding your data and assessing its quality.

How is your data organized in your system? Consider the details that matter to your business - for example, when is a product considered sold? Is it when the product is picked, shipped or invoiced? This definition can vary between industries, and it's often a strategic decision made by each company.

What can go wrong with historical sales data?

Many questions arise when ensuring the completeness of historical data. This goes beyond just collecting some sales numbers from some location in a certain period. Make sure you identify the missing values and the outliers and deal with them. Use techniques that fit best for you.

A common technique of dealing with missing values is interpolation, which means that you find the missing value by averaging the surrounding values.

The outliers are these extreme values that don't fit in the product pattern. The Demand Planning app has a function that helps users deal with outliers, I recommend always let users check the extreme outliers. Outliers tell a story and may reveal interesting secrets. Ask the magical question - **why** is this value so extreme?

And maybe you think - what can possibly be wrong with the historical sales data? It comes right out of my ERP system! Well, a lot can go wrong - I've put together some situations I could think of, there may be several. Feel free to add those in the comments section of this article.

<u>System migration or changes in ERP - CRM systems</u>
Historical data might be lost, corrupted, or inconsistently transferred during system migrations
<u>Missing or delayed data entry</u>
Sales data may not be recorded on time or at all due to manual entry errors, system downtime, or syncing issues.
<u>Product lifecycle - new, discontinued, seasonal SKUs</u>
- New SKUs may not have historical data yet - consider establishing a strategy for working with new products. I will share some thoughts about working with new products in a different article. - Discontinued or seasonal products may have large gaps in their sales history, which will make sense - identify those products when cleaning your data.
<u>Store or warehouse openings or closures</u>
The newly opened location will not have sales data, while the closed location may create a sudden drop-off in the datasets.

<u>Stockouts or supply chain disruptions</u>
No sales may have occurred during a period of stockouts, not due to lack of demand, but lack of availability, and sales data may falsely appear as zero demand. If you have knowledge of stockouts or supply chain disruptions, correct the data; this will add value to your forecast.
<u>Returns or cancellations not properly logged</u>
Sales numbers may be inflated or deflated if returns or cancellations aren't consistently reflected. If so, you might want to review your return process.
<u>Promotions or discounts not tracked</u>
Special campaigns or promotions may affect demand by being unrecorded, leading to anomalies
<u>Incorrect master data</u>
Errors in master data, such as wrong location codes or SKU IDs, can prevent proper matching and aggregation of sales history.

Now that we have been talking about the strategies of working with data, it's time to get practical and see how we manage data import in the Demand Planning app.

Demand Planning data import

In this section, I will first cover the system requirements, the connection setup with Dynamics 365 Supply Chain Management, and the creation of new tables, which will allow you to import custom data.

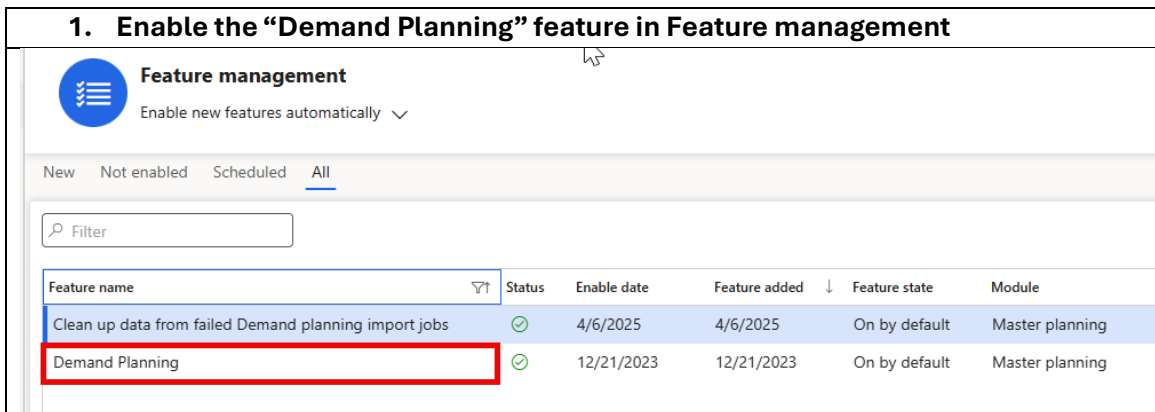
Finally, we take a look at the steps of creating an import profile in the Demand Planning app.

System requirements

- Use version 10.0.36 or later.
- Avoid cloud-hosted Supply Chain Management environments – **Tier2+ only**.
- Stay within the same Azure tenant.
- Get premium Supply Chain licenses and assign them to users in PPAC.

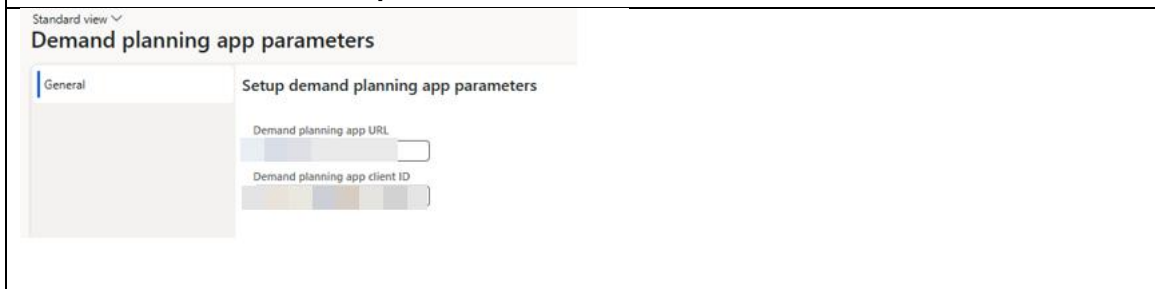
Connect Dynamics 365 Supply Chain Management with Demand Planning

1. Enable the “Demand Planning” feature in Feature management



Feature name	Status	Enable date	Feature added	Feature state	Module
Clean up data from failed Demand planning import jobs	✓	4/6/2025	4/6/2025	On by default	Master planning
Demand Planning	✓	12/21/2023	12/21/2023	On by default	Master planning

2. Enable Demand planning app parameters by navigating to System administration setup



Standard view ▾

Demand planning app parameters

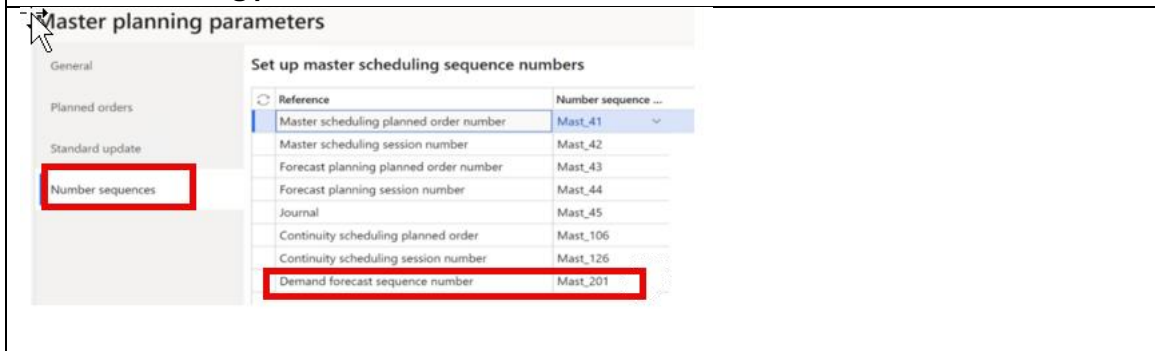
General

Setup demand planning app parameters

Demand planning app URL

Demand planning app client ID

3. Add sequence number by navigating to Master planning parameters and activating preallocations



Master planning parameters

General

Set up master scheduling sequence numbers

Reference	Number sequence ...
Master scheduling planned order number	Mast_41
Master scheduling session number	Mast_42
Forecast planning planned order number	Mast_43
Forecast planning session number	Mast_44
Journal	Mast_45
Continuity scheduling planned order	Mast_106
Continuity scheduling session number	Mast_126
Demand forecast sequence number	Mast_201

Mast_201 : Mast_201

Company USMF

Segments

+ Add Remove Move up Move down

Segment	Value	Length
Company	USMF	4
Constant	-	1
Alphanumeric	#####	7

Format
USMF-#####

References

General

SETUP

In use ☐ No ☐ Manual ☐ No ☐ ALLOW USER CHANGES

Stopped ☐ No ☐ Continuous ☐ No ☐ To a lower number ☐ No

To a higher number ☐ No

NUMBER ALLOCATION

Smallest 1 #999999 Next 1

Performance

Preallocation - You should increase the preallocation quantity according to your regulatory requirement for better performance. The functionality preallocate numbers in NumberSequenceList for continuous number sequences and in memory of the computer for have gap in the numbers. You can use the Preallocation functionality for continuous and non-continuous number sequences. You must specify how many numbers every tenth time a number is required. This functionality is very important for the general speed of the system. This is because selecting numbers from the database is considered.

ACTIVATE PREALLOCATION

Preallocation ☒ Yes ☐ No

ALLOCATED NUMBERS

Quantity of numbers 500 Next Largest

4. Refresh your entity list post-setup

Data import/export framework parameters

General

Entity settings

Bring your own database

Compatibility options

Advanced entity configuration settings

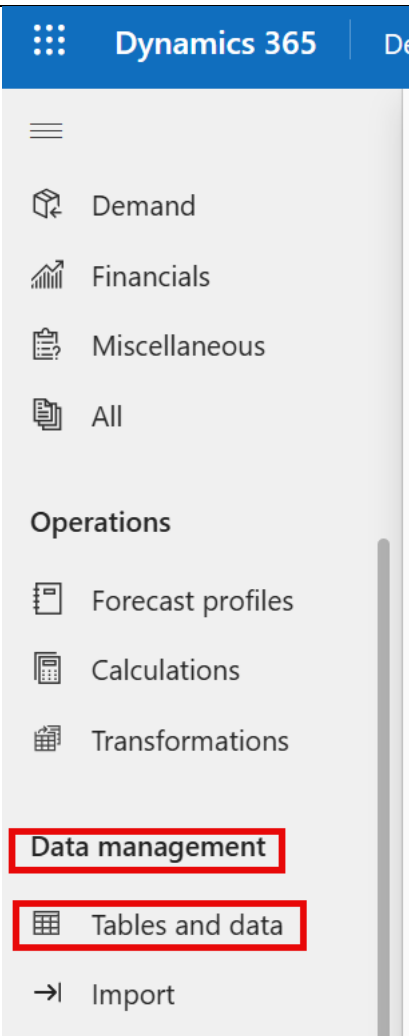
DATA IMPORT/EXPORT FRAMEWORK REQUIRES ENTITY LIST IS UP-TO-DATE. CLICK BUTTON BELOW TO REFRESH

CONFIGURE PROCESSING OPTIONS FOR ENTITIES TO OPTIMIZE DATA IMPORT/EXPORT OPERATIONS

Create a custom table in the Demand Planning app (if needed)

The Demand Planning app is designed to be flexible and allows you to import custom data. Before importing the data, you need to create tables in the application that reflect your data set.

1. Navigate to Data management – Tables and data



The screenshot shows the Dynamics 365 navigation pane. The 'Data management' option is highlighted with a red box, and the 'Tables and data' option is also highlighted with a red box. The navigation pane includes sections for Demand, Financials, Miscellaneous, All, Operations, and Import.

2. Click “New” to create a new table, and in the “General” tab, add a name.

New Table - Unsaved

General Columns Relationships Data

Name

* Demo table

Owner

*  Mihaela Remme (Available) ×

Key

Table is not created. Please create a ta

3. Add columns that reflect your data setGeneral **Columns** Relationships Data

+ New Column

☐ Name ↑ ↓	Data Type ↓	Is System ↓	Cr
☐ Country/Region	String	No	3/1
☐ Date	DateTime	No	3/1
☐ Inflation	Decimal	No	3/1

4. Return to General tab and add key to your table to ensure that no duplicates are allowed**Key**

Country/Region, Date ↓

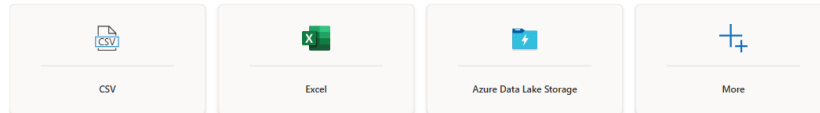
☒ Country/Region☒ Date☐ Inflation

Create an import profile in the Demand Planning app

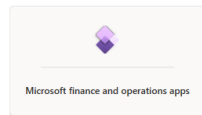
1. Navigate to Import under Data management
2. Click New
3. Select Microsoft Finance and Operations apps under other providers

Select data provider

Power Query Providers



Other providers



4. Enter a Name for the import profile.
5. Click Next

Create new import profile

Get started

This provider helps you retrieve data from Microsoft finance and operation apps such as Dynamics 365 Supply Chain Management. Retrieved data is accessible in Dynamics 365 Demand Planning and lets you perform demand forecasts and related business processes.

Name *

Description

Owner *

6. Enter the URL of the Dynamics 365 Supply Chain Management instance you want to connect to.
7. Click Next.

Create new import profile

Configure connection

Connect to datasource.

Microsoft finance and operations apps

Connection credentials

Connection URL *

Connection URL is required.

8. Consider selecting or deselecting the entities for synchronization
9. Click Next

Create new import profile

Get started

Configure connection

Entity selection

Review and finish

Entity selection

Please select which entities you want to import.

+ Add

Edit

Delete

Source	Target	Synchronize	Delete data
Historical Sales Demand Entity	Historical Demand	☒	☐
Demand planning picking list journal entries	Historical Demand	☐	☐
Distinct products or product variants	Product	☒	☐
Demand planning warehouses	Warehouse Location	☒	☐
Demand planning sites	Site Location	☐	☐
Demand planning customer accounts	Customer Accounts	☒	☐
Legal entities	Legal Entity	☒	☐

10. Click Review and finish.

Create new import profile

Get started

Configure connection

Entity selection

Review and finish

Name *

Description

Connection

Microsoft finance and operations apps

Connection credentials

Connection URL *

Entity selection

Source	Target	Synchronize	Delet...
Historical Sales D...	Historical Demand	☐	☐
Demand plannin...	Historical Demand	☐	☐
Distinct products...	Product	☐	☐
Demand plannin...	Warehouse Locat...	☐	☐
Demand plannin...	Site Location	☐	☐
Demand plannin...	Customer Accou...	☐	☐
Legal entities	Legal Entity	☐	☐

Back

Review and finish

Cancel

In this article, I've shared insight about the importance of data quality in the context of forecasting using AI and ML models in the Demand Planning app for Dynamics 365 Supply Chain management. The takeaways are based on hands-on experience from the past 16 years, along with a closer look at how to create import profiles within the app.

In the next article, we'll focus on transforming raw data into time series using the *Transformation* feature in the Demand Planning app.

Stay tuned as we explore how technology can turn forecasting into a strategic advantage.